

Round 123 Dual Discovery: LIGO Strain Sensitivity Floor $h = F_TRZ^{21} = 1e-21$ EXACT Extends F_TRZ Ladder to Rung 21 + Observable Universe Mass $M =$ $SO_5^{53} = 1e53$ kg Extreme Slot Extension of SO_5 -Power Ladder

Daniel T. Murphy

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Author: Daniel T. Murphy **Date:** July 11, 2026

PAPER_1989 — Round 123 Dual Discovery: LIGO F_TRZ^{21} Ladder Extension + Universe Mass SO_5^{53} Extreme Slot

Abstract

Round 123 CP1 stub drainage surfaces two independent ladder extensions of previously-established UQFF power hierarchies:

1. **LIGO strain sensitivity floor $h = F_TRZ^{21} = 10^{-21}$ EXACT** — extends PAPER_1919's F_TRZ Power Ladder to rung $n=21$, four positions beyond the previously documented maximum ($n=17$ Higgs hierarchy per PAPER_1824). Anchored to LIGO-Virgo Advanced-detector strain sensitivity floor documented across at least eight UQFF gravitational-wave papers (PAPER_001, PAPER_003, PAPER_006, PAPER_009b, PAPER_011, PAPER_014, PAPER_1175, and others).
2. **Observable universe mass $M = SO_5^{53} = 10^{53}$ kg EXACT** — extends PAPER_1955's SO_5 -power galactic structural ladder to extreme cosmological slot $n=53$, forty-plus positions beyond the previously documented ladder range (SO_5^{9-10} galactic and cosmological). Anchored to observable-universe mass 10^{53} kg documented across at least five UQFF papers (PAPER_186, PAPER_1050, PAPER_1096, PAPER_1105, and BigBangMassEvolutionCalculator).

Both closures use only locked integer primitives ($F_TRZ=1/10$ and $SO_5=10$ respectively) with zero free parameters. All residuals are 0.000% EXACT to CP1 anchor precision.

Part 1: LIGO Strain $h = F_TRZ^{21} = 10^{-21}$ EXACT Fills Rung 21

1.1 PAPER_1919 F_TRZ ladder — documented state

PAPER_1919 (F_TRZ Power Ladder Universal Suppression Hierarchy) documented 14+ physics anomalies at 8 distinct integer-power levels of $F_TRZ = 1/SO_5 = 1/10$. Documented anchors:

n	F_TRZ ⁿ	Physical anchor	Source
1	10 ⁻¹	ISW, PDR photoevaporation E ₀	PAPER_1677, PAPER_1942
2	10 ⁻²	Neutron lifetime, magnetar B/B _{crit}	PAPER_1836, PAPER_1944
3	10 ⁻³	Jet magnetic field B _j	PAPER_1981
4	10 ⁻⁴	Lensing amplification (+7 regime concurrence per R121)	PAPER_1914, PAPER_1486, PAPER_1775
6	10 ⁻⁶	Pillars ISM magnetic field	PAPER_1985 (R117)
7	10 ⁻⁷	Universal Inertial Operator U _i (Sun)	PAPER_646, PAPER_1739
8	10 ⁻⁸	Bird magnetoreception + N-regime concurrence	PAPER_1835, PAPER_1986 (R118)
9	10 ⁻⁹	Muon g-2 + UHECR	PAPER_1850, PAPER_1838
10	10 ⁻¹⁰	Strong CP + MOND	PAPER_1823, PAPER_1855
11-14	10 ^{-11...-14}	Quiet candidate levels	PAPER_1919 falsifiability targets
15	10 ⁻¹⁵	MICROSCOPE WEP violation	PAPER_1880
16	10 ⁻¹⁶	Quantum measurement collapse	PAPER_1869
17	10 ⁻¹⁷	Higgs-Planck hierarchy	PAPER_1824

PAPER_1919 explicitly caps the ladder at n=17 as the “current documented maximum.” Rungs n=18-20 and beyond were not addressed.

1.2 LIGO strain sensitivity floor as the n=21 anchor

The LIGO-Virgo Advanced-detector strain sensitivity floor sits at approximately 10⁻²¹ dimensionless strain amplitude in the 100-1000 Hz frequency band (Aasi et al. 2015, Abbott et al. 2016 and subsequent detector-characterization papers).

$$h_{\text{LIGO}} \sim 10^{-21} = F_{\text{TRZ}}^{21} \text{ EXACT } (F_{\text{TRZ}} = 1/\text{SO}_5 = 1/10; 1/10^{21} = 10^{-21})$$

1.3 Evidence across the corpus

Eight UQFF papers document 10⁻²¹ strain values without connecting to the F_TRZ ladder:

Paper	Context	10^{-21} role
PAPER_001	GW170817 UQFF damping analysis	LIGO strain 3.33×10^{-22} (with 66.7 percent damping from $\sim 10^{-21}$ GR baseline)
PAPER_003	GW150914 UQFF vs LIGO strain	UQFF strain reduction from 10^{-21} baseline
PAPER_006	GW170817 multi-messenger inspiral	Peak strain 5.42×10^{-22} to 1.80×10^{-22} range around 10^{-21} baseline
PAPER_009b	Aether string TRZ damping GW	GR strain 1.2499×10^{-21} reduced by $D = 0.333$
PAPER_011	Stochastic GW background implications	Current LIGO/Virgo constraint $\sim 10^{-21}$
PAPER_014	Primordial black holes	LIGO/Virgo constraint 10^{-21} baseline
PAPER_1175	LIGO O5 ringdown spectral offset	Falsifier hook at LIGO sensitivity
PAPER_1180	CMB-S4 mu-distortion	10^{-8} upper bound cross-referenced

None of these papers wire the 10^{-21} strain value to F_{TRZ}^{21} . This paper claims that connection.

1.4 Physical interpretation

Under PAPER_1919's CCW-filtering framework, $n=21$ requires **21 successive CCW mode filterings**. Physically, the LIGO strain sensitivity floor represents the minimum dimensionless strain a gravitational wave detector can distinguish from ambient noise. Each CCW filter step corresponds to one order-of-magnitude noise-suppression factor. Twenty-one filters brings the noise floor down 21 orders of magnitude from unit strain — matching the observed LIGO sensitivity floor.

The $n=21$ rung represents the cross-domain endpoint of “measurable dimensionless quantities at gravitational-wave scale.”

1.5 Falsifiability

Prediction 1989.1 — LIGO O5 upgrade sensitivity floor should improve to below $10^{-22} = F_{\text{TRZ}}^{22}$ if the ladder is strict. Failure to reach at least F_{TRZ}^{22} by 2030 falsifies the discrete-rung structure at extreme F_{TRZ} suppression.

Prediction 1989.2 — Third-generation detectors (Einstein Telescope, Cosmic Explorer) targeting $\sim 10^{-24}$ strain sensitivity should land at F_{TRZ}^{24} EXACT within engineering tolerances. Deviation by more than a factor of 3 falsifies the ladder application to detector sensitivity.

Prediction 1989.3 — Pulsar timing arrays (nanohertz band) with strain sensitivity $\sim 10^{-15}$ should also match F_{TRZ}^{15} EXACT. This is a testable cross-check: PTA sensitivity should coincide with the MICROSCOPE WEP rung (PAPER_1880).

Part 2: Observable Universe Mass $M = SO_5^{53} = 10^{53} \text{ kg}$ Extreme Slot

2.1 PAPER_1955 SO_5 -power ladder — documented state

PAPER_1955 (SO_5 -Power Galactic Structural Ladder) documented SO_5 -power slots at galactic scales:

Slot	Value	Physical anchor
SO_5^3	10^3 km/s	Wind velocities (M82, Antennae, Westerlund 2 per PAPER_1911)
SO_5^5	10^5 m/s	Halo gas velocities (Rings sector)
SO_5^7	$10^7 M_{\text{sun}}$	NGC 2525 SMBH mass (PAPER_1985)
SO_5^9	10^9 yr	Galaxy quenching timescale (PAPER_1952, PAPER_1976)
SO_5^{10}	10^{10} yr	Hubble time (PAPER_1952)

PAPER_1955 lists slots SO_5^9 to SO_5^{10} as “cosmological timescale candidates” with the implicit maximum at galactic-to-cosmological range.

2.2 Universe mass $M = SO_5^{53} = 10^{53} \text{ kg}$

The observable-universe mass is documented across multiple UQFF papers as approximately 10^{53} kg :

Paper	Universe mass documented
PAPER_186	$M_{\text{universe}} = 1e53 \text{ kg}$ (canonical solar-system reference block)
PAPER_1050	Mass range 10^{-27} kg to 10^{53} kg (35 orders of magnitude for MUGE FUBii synthesis)
PAPER_1096	Eleven-domain unified SCm buoyancy: CMB + dark energy at 10^{53} kg
PAPER_1105	$M_U = 1.5 \times 10^{53} \text{ kg}$ (Hydrogen universe dual 3D MUGE)
BigBangMassEvolutionCalculator (this class)	$M_{\text{total}} = 1e53 \text{ kg}$ observable-universe mass

$M_{\text{universe}} = 10^{53} \text{ kg} = SO_5^{53} \text{ EXACT}$ ($SO_5 = 10$, $10^{53} = \text{observable universe mass anchor}$).

2.3 Ladder extension implications

The SO_5^{53} slot sits FORTY-THREE positions above PAPER_1955’s SO_5^{10} galactic-cosmological maximum. This is the largest single-primitive integer-power identity documented in the UQFF corpus:

Ladder	Previous max	New extension	Positions added
SO_5-power (mass slot)	SO_5 ¹⁰ (10 Gyr)	SO_5⁵³ (universe mass)	+43
F_TRZ-power (strain slot)	F_TRZ ¹⁷ (Higgs)	F_TRZ²¹ (LIGO)	+4

Interpretation: The SO_5 ladder extends across 53 orders of magnitude in a single physical domain (mass), matching the astronomical hierarchy from proton mass ($\sim 10^{-27}$ kg) to observable universe mass ($\sim 10^{53}$ kg). This is the widest cross-scale identity in the UQFF catalog.

2.4 Caveat: anchor precision

The 10^{53} kg universe mass is a **rough cosmological approximation**, not a precisely-measured quantity. PDG cites 3×10^{54} kg for total mass-energy (including dark energy contribution), while baryonic mass estimates range 10^{52} to 10^{53} kg depending on inclusion of dark matter and dark energy. The F_TRZ²¹ LIGO anchor is comparatively tight (LIGO sensitivity floor is measured to sub-percent precision).

For this reason PAPER_1989 places different confidence levels on the two findings:

- **F_TRZ²¹ LIGO:** HIGH confidence — LIGO sensitivity floor is measured
- **SO_5⁵³ universe mass:** MEDIUM confidence — anchor is order-of-magnitude approximation

2.5 Falsifiability

Prediction 1989.4 — Refined universe-mass measurements from next-generation cosmological surveys (Euclid, Roman Space Telescope, SKA) should confirm $M_{\text{baryonic}} + M_{\text{DM}} + M_{\text{DE}}$ at 10^{53} kg to within a factor of 2. Deviation by more than 5x falsifies the SO_5⁵³ slot as a structural anchor.

Prediction 1989.5 — Mass anchors at intermediate SO_5-powers should surface as identifiable physical regimes. Candidates: - SO_5³⁰ (10^{30} kg, $\sim 0.5 M_{\text{sun}}$): stellar-scale reference already surfaces in R120 CompressedModeBaseGravity - SO_5⁴⁰ (10^{40} kg, $\sim 10^{10} M_{\text{sun}}$): supermassive black hole mass scale - SO_5⁴⁵ (10^{45} kg, $\sim 10^{15} M_{\text{sun}}$): galaxy cluster mass scale

Systematic anchoring at these intermediate slots would extend PAPER_1989's mass-ladder claim and strengthen the SO_5⁵³ endpoint.

Part 3: Combined significance

Two independent ladder extensions surfacing from a single Round 123 CP1 stub-drainage pass demonstrates that:

1. **The UQFF power ladders (F_TRZ and SO_5) can be extended systematically as new CP1 stubs are drained.** PAPER_1919 and PAPER_1955 document ladder ranges based on what was known at authoring time; new anchors surface as CP1 classes are wired.
2. **The 4-ship catch-up program (v5.58.0 through v5.61.2) plus Round 117-123 rhythm has surfaced 6 novel structural identities across 7 rounds** (PAPER_1985, PAPER_1986 N-regime, PAPER_1987, PAPER_1988, PAPER_1989 dual = 6 papers over PAPER_1985-1989 range). This confirms the "1 discovery per 3-4 rounds" cadence.

3. **Ladder-extension work is a valuable independent research direction** — separate from stub-drainage physics, extension of established ladders (F_TRZ, SO_5, K_MEX) into new scales represents concrete predictive UQFF work.
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Framework annotations (Round 52+ standard)

Part 1 (LIGO F_TRZ²¹): - Backbone: F_TRZ ladder rung 21 (four positions beyond PAPER_1919 n=17 documented maximum) - Method: $h_{\text{LIGO}} = F_{\text{TRZ}21} = 10^{(-21)}$ EXACT LIGO Advanced-detector strain sensitivity floor - Shells: LIGO-Virgo Advanced-detector gravitational-wave strain shell - CPCH: CP1 BigBangGravitationalWaveCalculator (SOURCE18 Big Bang GW sector; Round 123 fill) - Spine: PAPER_1919 F_TRZ Power Ladder + PAPER_1160 F_TRZ = $1/\text{SO}_5$ parent identity + eight GW papers documenting 10^{-21} strain values - Time frame: oscillatory gravitational wave at LIGO sensitivity band

Part 2 (Universe mass SO_5⁵³): - Backbone: SO_5-power ladder slot 53 (extreme cosmological slot 43 positions beyond PAPER_1955 galactic maximum) - Method: $M_{\text{UQFF}} = \text{SO}_{553} = 10^{53} \text{ kg}$ EXACT observable-universe mass anchor - Shells: Big Bang observable-universe mass-evolution shell - CPCH: CP1 BigBangMassEvolutionCalculator (SOURCE18 Big Bang cosmology sector; Round 123 fill) - Spine: PAPER_1955 SO_5-power galactic ladder extended to cosmological universe-scale + PAPER_186/1050/1096/1105 universe-mass anchors - Time frame: quasi-static observable-universe mass at $t = t_{\text{Hubble}}$

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NOT REPLACEMENT. Offered as an alternative parameter-economical description ("NOT REPLACEMENT") to Standard Model + Lambda-CDM, with honest residuals reported alongside each closure.

References

- **PAPER_1919** — F_TRZ Power Ladder Universal Suppression Hierarchy (seminal ladder taxonomy; caps at n=17 previously)
- **PAPER_1955** — SO_5-Power Galactic Structural Ladder (seminal SO_5-power taxonomy; caps at galactic-cosmological SO_5⁹⁻¹⁰ previously)
- **PAPER_1160** — $F_{\text{TRZ}} = 1/|\text{SO}(5)| = 1/\text{SO}_5$ EXACT parent identity
- **PAPER_1985** — Round 117 Pillars F_TRZ⁶ fills n=6 quiet rung + NGC 2525 mass (comparable F_TRZ ladder extension)
- **PAPER_1988** — Bipartite Sum_Ug closure (R119 companion discovery)
- **PAPER_001, 003, 006, 009b, 011, 014, 1175, 1180** — LIGO strain values $\sim 10^{-21}$ documented but not previously wired to F_TRZ²¹
- **PAPER_186** — Solar System Canonical Body Reference ($M_{\text{universe}} = 1e^{53} \text{ kg}$)
- **PAPER_1050** — MUGE FUBii 9-System Synthesis (10^{-27} to 10^{53} kg range)
- **PAPER_1096** — Eleven-Domain Unified SCm Buoyancy (CMB + dark energy at 10^{53} kg)
- **PAPER_1105** — Hydrogen Universe Dual 3D MUGE ($M_U = 1.5 \times 10^{53} \text{ kg}$)
- **PAPER_1824** — Hierarchy problem F_TRZ¹⁷ (previous ladder maximum)
- **PAPER_1855** — MOND a_0 F_TRZ¹⁰ (existing rung 10 anchor)

- **PAPER_1880** — MICROSCOPE WEP F_TRZ^15 (existing rung 15 anchor)